

Dropout Classification for a mobile Diabetes Prevention Program via Topic Modelling

Introduction

Approximately 115.2 million American adults, more than 1 out of 3, have prediabetes. Of those with prediabetes, 80% are not aware they have this condition [1]. The Diabetes Prevention Program (DPP) trial [2] and other large trials [3, 4, 5] demonstrated that a moderate amount of weight loss achieved by reducing total caloric intake and increasing physical activity can prevent or delay the onset of type 2 diabetes. However, these prevention programs that involve intensive in-person counseling are expensive to implement and sustain in clinical or community settings over extended periods of time.

The rapid rise in sophisticated digital tools (i.e., mobile applications, (wireless activity trackers, social media) provides an opportunity to transform the way health care teams deliver a diabetes prevention and weight loss program. These digital technologies are inexpensive and widely available and can reach large numbers of adults with prediabetes and obesity. However, despite their popularity and availability, the vast majority of individuals will stop using them within weeks because of the lack of a systematic behavior change approach (i.e. setting personalized short-term and long-term goals). Therefore, there is an urgent need to reverse this trend and develop interventions that combine the best elements of comprehensive in person diabetes prevention programs with inexpensive and easy to use digital technologies to address the epidemic of prediabetes and obesity effectively and sustainably, not just to the point of achieving short term goals, but for a lifetime.

In order to take advantage of these interventions, it is important to minimize the amount of participant dropout and help guide the implementation of them. BERTopic, a topic modeling model, will be used in order to classify reasons for dropout from the study in order to better understand the challenges potential participants may be facing.

Dataset

For the purpose of this study, the intended dataset does not currently have enough datapoints to run BERTopic on it. We will substitute a kaggle dataset of news headlines. These headlines are similar to the intended data topic in that they are of similar length, and for this particular dataset, all are financial headlines. This is applicable because our intended dataset also falls under a single overall topic, dropout, and we intend to find subtopic clusters within this topic. The dataset consists of 17252 headlines with 3 columns; the count of the headline, the title of the headline, and a binary classification label determining a negative or neutral/positive tone. For the purpose of our BERTopic model, only the second column is necessary.

Metric	Value
Total documents	17,252
Average length	11.1 words
Median length	10 words
Range	1–51 words

The average length of document is 11.1. This indicates that the reasoning behind choosing this dataset is valid, as this will test the model on short documents.

BERTopic

BERTopic is a topic modelling model that utilizes pre-trained sentence embedders compared to bag of words tokenization. This makes it an attractive choice over other models such as Latent Dirichlet Allocation and Non-Negative Matrix Factorization due to its ability to classify short documents. BERTopic is broken down into 5 components; embedding, dimensionality reduction, clustering, tokenization, and weighting.

For the purpose of this dataset, Sentence-BERT (SBERT) will be used as the sentence embedder. This embedder embeds the context of entire sentences rather than of individual words, which is useful in this case as we are attempting to cluster together documents that consist of sentences. There are various encoders that can be used through this embedder, all-Mini-LM-L6-v2 is the encoder that will be used. This is a common English language model that will break each document down into 384 dimensions.

After the documents are embedded, they are dimensionally reduced. For BERTopic, non-linear dimensionality reduction is preferred as semantics are non-linear in relation. Unifold Manifold Approximation and Projection (UMAP) is used for this. UMAP starts out by constructing a high dimensional graph. It calculates a local radius for each point based on the distance to the n th nearest neighbor. N in this case is a chosen parameter. In order to transform this to a lower dimensionality, we use simplices, which are just ways to build a k dimensional object. A zero simplex is a point, a one simplex is a line, a two simplex is a triangle, a three simplex is a tetrahedron, and so on. For the purpose of UMAP only zero and one simplices are used. Overlapping radii are given a one simplex. When comparing the high and low dimensional spaces, all zero simplices, or points, are shared between them. If we look at a set of all possible one simplices, we can treat these as Bernoulli variables, in that the one simplex either exists or it doesn't, allowing us to use cross entropy.

$$\sum_{e \in E} w_h(e) \log \left(\frac{w_h(e)}{w_l(e)} \right) + (1 - w_h(e)) \log \left(\frac{1 - w_h(e)}{1 - w_l(e)} \right)$$

This equation represents the cross entropy equation, where the front half attempts to pull together points that were neighbors in the higher dimension, and the second half of the equation tries to push apart those that were not.

After the data is dimensionally reduced, it is next clustered. The default clustering method is Hierarchically Density Based Spatial Clustering Applications with Noise, or HDBSCAN. This starts by using KNN, and assigning a core distance between the point and its k th neighbor.

We then convert the distances into mutual reachability distance. The mutual reachability distance between two cores, when spanning the radii to the k th neighbor from both cores, is just the distance between the two cores. When the distance between two cores doesn't span the radii of one of the cores, the mutual reachability distance is from the core to the larger radii.

To build the tree, we set a threshold. We assign weights to each of the mutual reachability distance lines, and drop weights that are above that threshold. This eventually creates our tree. We can do this efficiently by building it one edge at a time, adding the lowest weight that connects the tree to a new point.

To build the cluster hierarchy, we sort all weights into ascending order, and then begin removing them into separate clusters, starting with the largest. Eventually, every point will be isolated.

Immediately after we split all the clusters, we recondense them back into larger clusters. We can set a minimum cluster size to ensure that we don't have hundreds of clusters consisting of individual points. We walk through each node split and determine if either node created falls under the minimum cluster size.

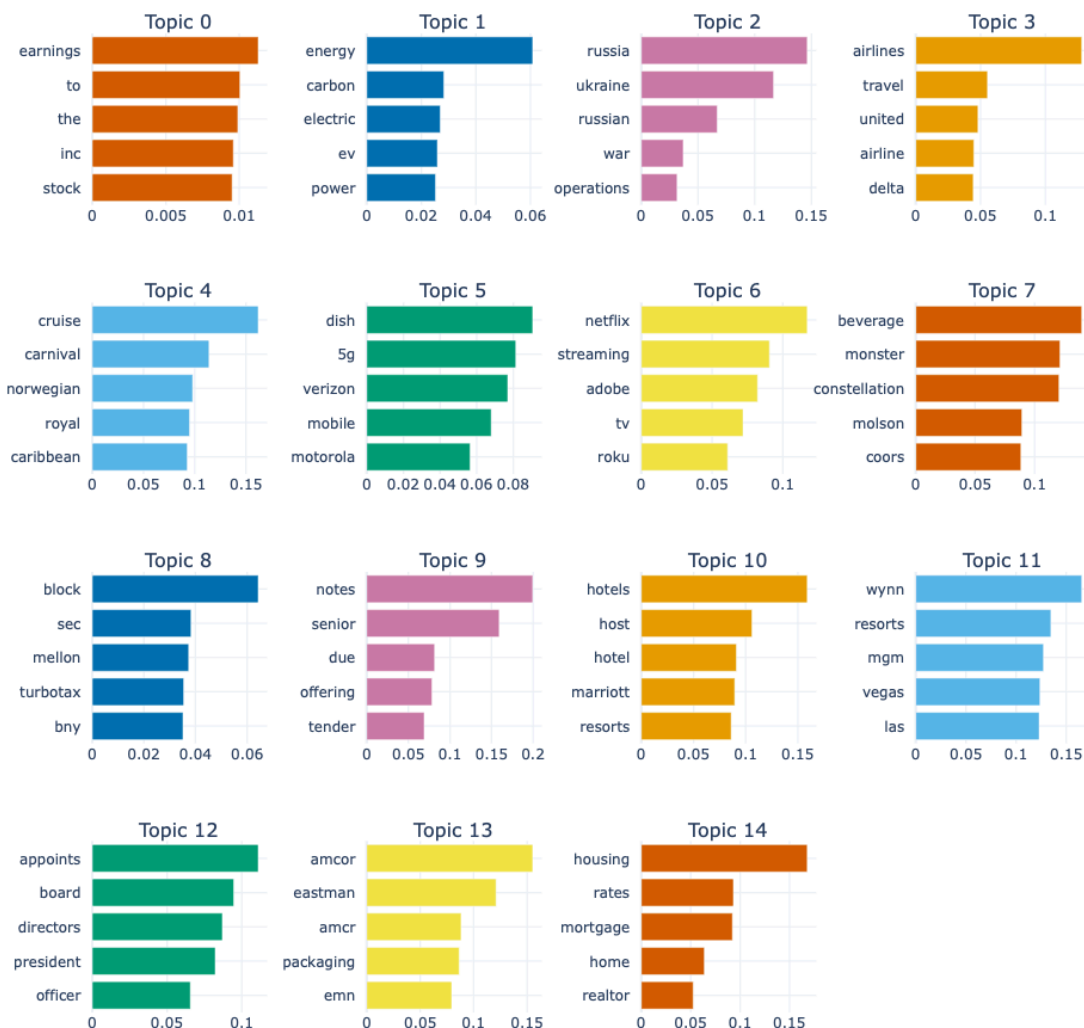
In order to extract clusters, we use a simple formula, 1 over the distance to determine a value λ . This λ value is assigned for multiple instances, λ birth and λ death for when a cluster forms and then when it splits, and λ_p for each point within a cluster. We do some summation between the points and births to determine the overall stability of the cluster, which basically is just to say that we are selecting the clusters with the largest area created from this summation formula.

Now that we have our clusters, we want to interpret them. In order to extract meaning from our embeddings, we actually need to go back to bag of words which we opted to not use in the beginning. In order to do this, we combine all documents within a cluster into a single document. We then run bag of words on each cluster individually. This gives us topic level word representations.

We can now run our class-based term frequency inverse document frequency model on the words. Rather than giving us the importance of words between documents, it now gives us the importance of words between clusters. The plus one within the logarithm keeps all of our values positive in this case. This half gives us our class based inverse document frequency representation, which we then multiply by our term frequency in order to get the score for each word in each class. This should give us a set of words that describe each topic.

Discussion

Topic Word Scores



Our topic model yielded 15 distinct topics. Looking at the output it is noticeable that a few topics have overlapping words, and some of the words are not necessarily useful for ascertaining topics. In cases like this, it may be useful to apply some stopwords removal during the tokenization phase. Additionally, tighter constraints could be used for some of the parameters during the construction of the overall BERTopic model in order to maintain more uniqueness between topics. Overall, this output is very promising when it comes to attempting to sort through topics that generally are within a single topic umbrella, as even though some of the topics display some overlap, the majority of topics are distinct contextually.

References

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